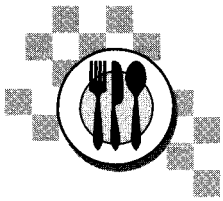
CHAPTER 9



Pest Control

About This Chapter

In this chapter you will learn:

1. Which pests (insects, rodents, and birds) cause the most problems in food establishments
2. How to prevent pest infestations through inspections and good housekeeping
3. How to detect pest infestations
4. Why pest control is so difficult but essential
5. How to control pests, using chemicals, physical and mechanical methods, biological methods, and integrated pest management
6. How to use pesticides safely

Introduction

This chapter will not make you an expert on pest control, but it will give you a basic understanding of how insects, rodents, and birds contaminate food supplies. Foodservice staff need to know about pests that can contaminate the food supply and how to control them. The number of species of insects, rodents, and birds that cause problems in the food supply is small, but they cost the food industry billions of dollars every year.

Food establishments may employ an outside exterminator. Also, one or more employees (depending on the size of the organization) may be trained to take care of pest control.

Tidiness and cleanliness help keep pests away from the facility. They make it easier to exterminate pests, more difficult for pests to get into the building, and harder for pests to find a place to live and reproduce. Insects and rodents will not be able to

find shelter or food if rubbish, food waste, and equipment that is no longer used are disposed of quickly. Enclosed areas under shelves, platforms, chutes, and ducts, and holes in walls and insulation, make good homes for pests, especially if debris collects in these areas.

INSECTS

Cockroaches

Cockroaches are the most common pests in food-processing plants and foodservice facilities throughout the world. These pests must be controlled because they carry and spread various diseases. Many carry the microorganisms that cause poliomyelitis and cholera.

Cockroaches spread microorganisms when they crawl on, bite, and chew food. They prefer high-carbohydrate (starchy or sweet) foods, but will eat any food, human waste, decaying food scraps, dead insects (including other cockroaches), shoe linings, paper, or wood. Cockroaches are most active in dark places and at night where humans are less likely to disturb them.

Each month, these pests lay small egg cases containing 15 to 40 eggs. Some cockroaches deposit the egg case in a hiding place to protect it. Shortly after they hatch, young cockroaches begin eating the same things the adults eat. Young cockroaches look like adults, except that they are smaller and do not have wings. They develop wings after they grow and shed their skin several times. Cockroaches may live over a year and mate several times.

Food sanitarians need to know the type of cockroach that has infested an establishment, so that they can use the right method to remove it. The three most common cockroach species in commercial establishments in the United States are the German cockroach, the American cockroach, and the Oriental cockroach. However, other cockroaches, such as the field cockroach (found on crops), are starting to cause problems in homes and food establishments in parts of the southern United States.

Species

German cockroach. The German cockroach is pale brown, with two dark-brown stripes behind the head. All adults have well-developed wings. The female carries the egg case behind the tip of her abdomen until the eggs hatch.

In food establishments, German cockroaches can infest the main processing or preparation rooms, storage areas, offices, and washroom facilities. They prefer to live in warm crevices near warm pipes, stoves, boilers, or other heating equipment, and do not usually live in basements. German cockroaches are common in restaurants and may be found anywhere in a room, from the floor to the ceiling.

American cockroach. This is the largest cockroach in the United States. Adults are reddish-brown, and the young are pale brown. The female hides the egg cases as soon as she produces them.

American cockroaches live in open, wet areas such as basements, sewers, drainage areas, and garbage areas, as well as storage rooms and loading docks. They prefer cool places that have larger crevices than the German cockroach prefers.

Oriental cockroach. The Oriental cockroach is shiny dark brown or black. The male has very short wings, and the female has no wings. Its young are pale brown. The female hides the egg cases soon after she produces them.

This species prefers similar areas to those preferred by the American cockroach. In food plants, they normally live in basements or damp storage areas.

Detection

Cockroaches live anywhere where food is processed, stored, prepared, or served. The insects hide and lay eggs in dark, warm, hard-to-clean areas. They like small spaces in and between equipment and under shelf liners. When cockroaches need food that is not in these areas or they are forced out by other cockroaches, they will come out into the light.

One of the easiest methods of checking for cockroaches is to walk into a dark production or storage area and then turn on the lights. Cockroaches produce a strong, oily odor from their glands. They also leave feces (small, black or brown, and spherical) everywhere they go.

Control

Cockroaches infest food establishments all year round, so all workers need to continuously use good sanitation and chemicals to keep them under control. Good sanitation is the most important form of control. Because these insects will eat almost anything, workers should keep all areas of the operation, including washroom facilities, tidy and clean.

Filling cracks in floors and walls with caulking or other sealants reduces the number of cockroaches. Spaces between large pieces of equipment and their bases or the floor should be sealed. These spaces provide an ideal home for pests. Cockroaches get into food establishments as cockroaches or as eggs in boxes, bags, raw foods, or other supplies. All deliveries should be examined with the removal of any insects or eggs. Cartons and boxes should be taken to the trash area as soon as the supplies are unpacked.

An outside pest controller may handle chemical control of pests, but a combined chemical and sanitation program within the organization can be more effective and cheaper. Cockroach control uses baits and bait stations, fungi, and sometimes nematodes (parasitic worms). Cockroaches become inactive at about 5°C (41°F), so refrigeration and freezing reduces their numbers.

The chemical amidinohydrozone can kill cockroaches that resist other poisons. An insecticide such as diazinon sprayed into hiding places works well if the pests have not developed a resistance to this compound. Pyrethrin-based insecticide is sometimes added to diazinon to force the insects out of hidden areas into the

sprayed area. Some form of diazinon can be used in spots, cracks, or crevices, but not in food-handling areas. Workers must follow the directions on the label when they use any insecticide.

Other Insects

The most common seasonal insects in foodservice and food-processing plants are houseflies and fruit flies.

Housefly. It lives all over the world and is an even greater health problem than the cockroach. It carries several types of pathogens to humans and their food, causing typhoid, dysentery, infantile diarrhea, streptococcal and staphylococcal infections, and other diseases.

Flies eat animal and human wastes and collect pathogens from the waste on their feet and wings and in their mouths and guts. When flies walk on food or eat food, they leave pathogens on it. Flies take in food as liquid, so they secrete saliva onto solid food to liquefy it before they suck it up. Fly saliva is loaded with bacteria that contaminate food and equipment.

Control of flies is challenging, because these pests can get into buildings through openings only slightly larger than the head of a pin. Flies normally stay close to the area where they hatch, but they are attracted to smells and decaying materials. Air currents often carry flies further than they would fly. Flies prefer to live in warm areas out of the wind, such as garbage can rims. Warm, damp, shaded decaying material provides an ideal place for housefly eggs to hatch and grow as fly larvae or maggots.

There are more houseflies during late summer and fall because the population builds rapidly during warm weather. Once adult flies enter buildings for food and shelter, they usually stay there. Flies are most active at temperatures between 12°C and 35°C (54–95°F). Below 6°C (43°F) they are very inactive, and below 5°C (41°F) they die within a few hours. Heat paralyzes flies at about 40°C (104°F) and kills them at 49°C (120°F).

The size of a housefly population is difficult to control, because they often breed away from food establishments in decaying material. The best ways to control the fly population are to prevent flies from entering processing, storage, preparation, and serving areas and to reduce their numbers within these areas.

Sanitarians can prevent flies from getting into food establishments by removing waste materials from food areas promptly and thoroughly. Air screens, mesh screens (at least 16 mesh is recommended by the U.S. Public Health Service), double doors, and self-closing doors help keep flies out. Staff should keep doors closed whenever possible and open them for the shortest possible amount of time for receiving or shipping.

To keep flies away from a food establishment, staff should store the outdoor garbage as far away from doors as possible. If the facility stores garbage inside, a wall should separate this area from other areas, and refrigeration should be used to reduce decay and fly activity. Garbage containers should have lids.

If flies do get into the facility, an electric fly trap attracts the adults to a blue light and kills them using an electric grid. Electric fly traps should run all day, and the catch basin should be cleaned daily. Aerosols, sprays, or fogs of chemicals such as pyrethrins can help control flies. But the results are temporary, and chemicals are restricted in food facilities. Therefore, keeping flies out and using fly traps are preferred methods of controlling flies.

Fruit fly. The fruit fly is smaller than the housefly and is also most abundant in late summer and fall. Adult fruit flies have red eyes and light-brown bodies. They are attracted to fruit, especially rotting fruit. These pests are not attracted to sewage or animal waste, so they carry fewer harmful bacteria.

The life cycle and feeding habits of fruit flies are similar to those of houseflies, except that these insects are attracted specifically to fruits. Fruit flies grow fastest in late summer and early fall when there are plenty of rotting plants and fruits. Fruit flies live about 1 month.

It is hard to get rid of fruit flies completely. Mesh screens and air screens keep them out of food establishments. When they do get in, electric traps help keep their numbers down. The best way to control these pests is to get rid of rotting fruits and fermenting foods quickly.

Other insect pests. In food-processing and foodservice operations, they include ants, beetles, and moths. These pests leave webs or holes in food and packaging materials, and they can be controlled by keeping the facility tidy and well ventilated, keeping storage areas cool and dry, and rotating stock.

Ants often build nests in walls, especially around heat sources, such as hot water pipes. Sponges saturated with syrup can act as bait to find out where ants are living and where to place insecticide. Because ants, beetles, and moths need only very small amounts of food, good sanitation and proper storage of food and supplies are essential to keep these pests away.

Silverfish and firebrats can live in cracks, baseboards, window and door frames, and between layers of pipe insulation. Because these pests like undisturbed areas, thorough and frequent cleaning usually keeps them away. Silverfish prefer a moist environment, e.g., basements and drains. The firebrat likes warmer places, such as around steam pipes and furnaces.

INSECT DESTRUCTION

Pesticides

Pests should be destroyed without using chemicals whenever possible, because pesticides can be dangerous. But sometimes pesticides are necessary if other methods do not work. Food producers should consult a professional pest control firm if they do not know how to apply pesticides properly. Only commercial pest controllers are allowed to apply restricted pesticides. Even when an exterminating firm is used, supervisors at the food establishment need a basic understanding of pests, insecticides, and regulations concerning use of chemicals.

Residual insecticides. These insecticides are effective for an extended period of time. Some residual insecticides are illegal in food areas. It is important not to contaminate food, equipment, utensils, and supplies that people will eat and workers will touch. Staff who use these chemicals should understand the terms on the product labels, how the product is allowed to be used, and effects the product may have.

Operators apply residual insecticides three ways:

- *Mass application* to floors, walls, and ceilings.
- *Spot treatment* in cracks and crevices where insects are likely to live. Spot areas may be on floors and walls or around equipment and supplies.
- *Crack and crevice treatment* for areas where insects hide or enter buildings, for example, expansion joints in building construction and between equipment and floors. It is important to treat these places, because the openings often lead to hollow walls or equipment legs and bases where pests like to live. Staff should also treat conduits, junction or switch boxes, and motor housings.

Nonresidual insecticides. These insecticides control insects only at the time they are applied. Operators apply nonresidual insecticides in two ways:

- *Contact treatment* means applying a liquid spray for an immediate insecticidal effect. Operators should use only this method when the spray is likely to touch the pests.
- *Space treatment* uses foggers, vapor dispensers, or aerosol devices to disperse insecticides in the air. This technique can control flying insects and crawling insects in the exposed area. Space spraying helps control the number of insects.

Operators may use nonresidual insecticides in food production areas only when food is not exposed. Examples of nonresidual insecticides are pyrethrins and pyrethroids. Aerosols are good for killing flying and exposed insects and are often released using a timer at a time when food production and contact does not occur.

The food industry uses fumigants to control insects that attack stored products. Fumigants are good at reaching hidden pests and are often used for space treatment on weekends when the plant is not operating, so that food and workers are not exposed. Ventilation machinery or fans help disperse fumigants properly.

The following chemicals are common fumigants for insects:

Phosphine: This insecticide comes in a permeable package or in pellets. The gas is very flammable, so it is important to follow the manufacturer's instructions for use and storage.

Methyl bromide: This nonflammable fumigant is widely used.

Ethylene oxide: Operators often mix this fumigant with carbon dioxide in a ratio of 1:9 (by weight) to reduce flammability and explosiveness. A professional pest controller has to apply this compound.

Other Chemicals for Controlling Insects

Baits use sugar or other foods to attract insects and an insecticide to kill them. Baits may be less convenient than other methods, but they can work well against ants and cockroaches in inaccessible areas and against outside fly populations. Baits are a poi-

sonous food, so it is important to use and store them carefully. Operators may scatter commercial dry granular baits over feeding surfaces every day, or as needed, to control insect populations. Granular fly baits work well outdoors. Liquid baits contain an insecticide in water with sugar, corn syrup, or molasses. Operators spray or sprinkle liquid bait on walls, ceilings, or floors where flies are found. They should use fly bait regularly during the summer months to control growth of insect populations.

Mechanical Methods

Mechanical devices to control insects do not work very well. Fly swatters spread insect carcasses and bacteria, so they should not be used in food-processing, storage, preparation, or sales areas. The best mechanical device for controlling insects is an air curtain. This prevents insects and dust from entering food establishments and also reduces cold-air loss from refrigerated facilities. Air curtains can be used even at large entrances used by loading trucks or large equipment. An air curtain is created by a fan that sweeps air down over door openings and is usually mounted outside and above door openings.

Insect light traps. One of the safest and best ways to control flies is to use insect light traps. The advantage of this technique is that it is not toxic.

Insect light traps conduct a high-voltage, low-amperage current across a grid (see Fig. 9.1). This creates a light source that attracts flies, but when they fly toward it, they are electrocuted. Some light traps have a "black light," which works at night, and a "blue light," which works in the daytime.

Managers should install insect light traps in food-processing plants and warehouses in the following stages:

Stage 1, interior perimeter: Place these units near shipping and receiving doors, employee entrances, and staff doors that lead to the outside or anywhere else where

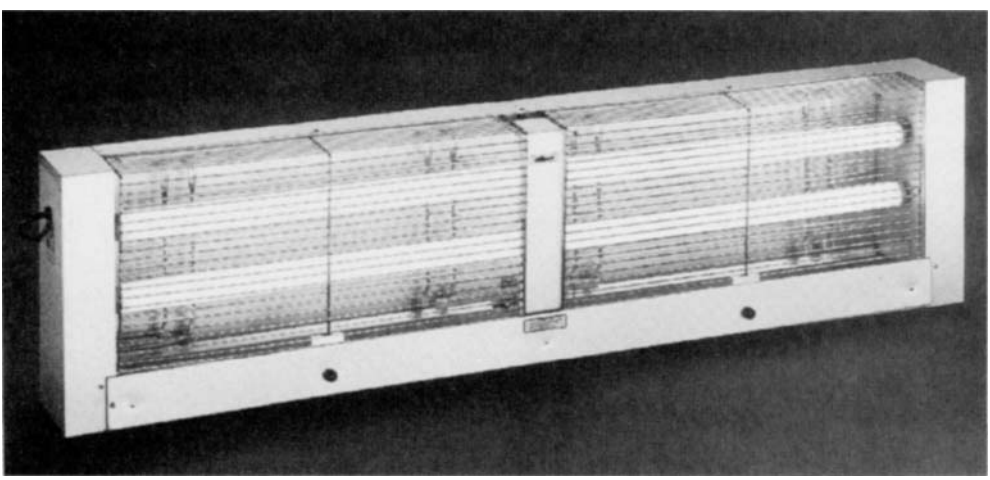


FIGURE 9.1. An insect light trap that attracts flies to the light source, subsequently electrocuting them. *(Courtesy of Dan Gilbert Industries, Inc.)*

flying insects may enter. Install units 3 to 8 meters inside doors, away from strong air currents and out of traffic areas where forklifts or other equipment may damage them.

Stage 2, interior: Place these units along the path that insects may follow to food-processing areas. Use units with wings within processing areas to prevent dead insects from falling on the floor or on processing equipment.

Stage 3, exterior perimeter: Protect covered docks, especially if they are used to store refuse. Install units between the insects and the entrances, but not directly at the entrances.

It is important to replace the light source every year for it to work properly. Operators need to place traps carefully so that they attract as many insects as possible, but not insects from outside. Staff should empty the pan that collects the electrocuted insects regularly so as not to attract dermestid beetles and pests that feed on dead insects.

Sticky traps. Sticky traps consist of sticky flypaper or a flat piece of plastic covered with a slow-drying adhesive. Yellow plastic strips with a sticky covering catch a wide variety of flying insects.

Biological Control

Biological control is a common part of integrated pest management (IPM) programs (discussed near the end of this chapter). Growth regulators may interrupt the life cycle of insects and prevent reproduction of mosquitoes, fleas, and other insects. Milled diatomaceous earth contains sharp microscopic particles that pierce the insects' wax coating so that they lose moisture and die. If particles of the shell get into the insect's body cavity, they interfere with digestion, reproduction, and respiration.

Hydroprene is an insect growth regulator (IGR) used to control cockroaches. The Environmental Protection Agency has approved its use in areas where food is present because it is relatively nontoxic and is not a pesticide. An IGR can disrupt the normal growth and development of immature cockroaches by deforming their wings and making them sterile.

Pheromone traps. These traps attract insects using insect sex hormones (pheromones) and have a trapping chamber to catch the insects. Some have a plastic funnel leading into a chamber that contains an insecticide strip. New products contain microencapsulated pheromones that release chemicals slowly over a long period of time. Pheromones are now available for various species, including fruit flies. Some food aromas may attract insects even better than pheromones.

RODENTS

Rodents such as rats and mice are difficult to control because they have highly developed senses of hearing, touch, and smell. These pests can identify new or unfamiliar objects in their environment and then protect themselves against these changes. Figure 9.2 shows some of the incredible things mice can do.

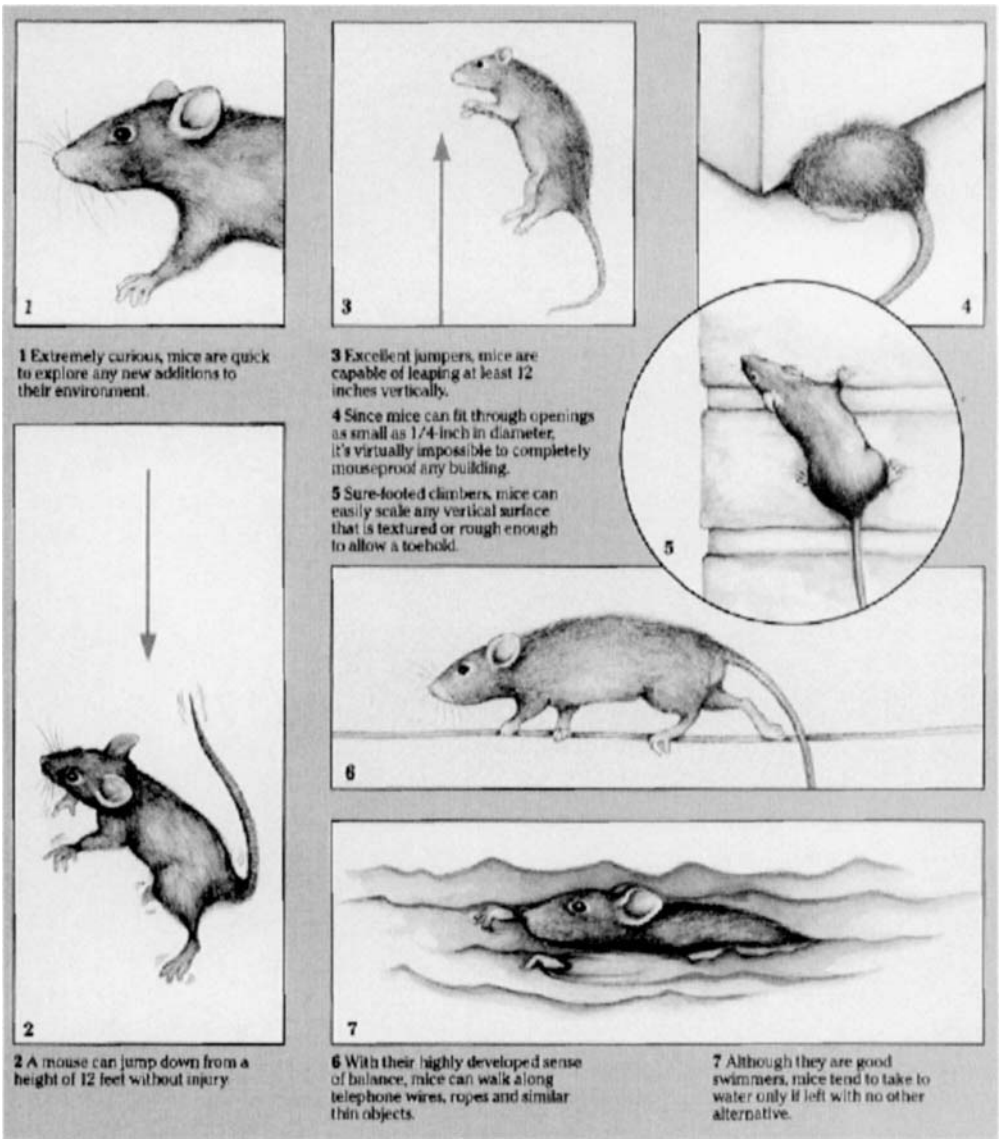


FIGURE 9.2. Mouse acrobatics.

Rats

Rats can enter buildings through openings as small as a quarter, climb vertical brick walls, and jump up to a meter (39 in) vertically and 1.2 meters (48 in) horizontally. These rodents are strong swimmers and can swim up through toilet bowl traps and floor drains. Rats have excellent balance and can even walk on suspended wires.

Rats are dangerous and destructive. The National Restaurant Association estimates that rodent damage costs $10 billion per year. This includes the costs of food they eat, food they contaminate, and structural damage to property, including dam-

age from fires started when rats gnaw on electrical wires. The serious health hazard from contaminated food, equipment, and utensils is even more serious than economic losses. Rats transmit such diseases as leptospirosis, murine typhus, and salmonellosis. One rat dropping contains several million harmful microorganisms. When droppings dry out and fall apart or are crushed, air movement within a room can carry particles to food.

The most common rat in the United States is the Norway rat, sometimes known as the sewer rat, barn rat, brown rat, or wharf rat. Norway rats are normally brown and are 18 to 25 cm (7.5–10.5 in) long, excluding the tail, have a blunt nose and thick-set body, and tend to live in burrows.

Roof rats live in the South and along the Pacific Coast and Hawaii. They prefer high places and have better coordination than Norway rats. Roof rats are black or slate-gray and are 16.5 to 20 cm (7–8.5 in) long, excluding the tail. Roof rats create nests in trees, vines, and other places above the ground.

Rats and mice instinctively avoid open spaces, especially if they are light colored. Therefore, a 1.5-m-wide (5-ft-wide) band of white gravel or granite chips around the outside of a building helps keep rodents out.

Mice

Mice can enter a building through a hole as small as a nickel. Mice are also good swimmers, can swim through floor drains and toilet bowl traps, and have excellent balance. Like rats, mice are filthy rodents and spread many diseases. The house mouse is found everywhere in the United States and has a body length of 6 to 9 cm (2.5–4 in). It has a small head and feet and large prominent ears.

Mice do not need to drink water, because they can survive on the water they release when they metabolize food. But they do drink liquids if they are available.

It is easy to carry mice into food premises in crates and cartons. They are easier to trap than rats because they are less careful. Metal and wood-base snap traps are normally effective. Operators may space several traps about 1 m (3 ft) apart. Mice usually accept new objects, such as traps, often after about 10 minutes. Sodium fluorosilicate and chlorophacinone are poisonous tracking powders that work well to control mice.

Finding Rodent Pests

Rats and mice tend to be inactive during daylight hours, so they are not always obvious. Droppings are an obvious sign of rodent infestation. Fresh droppings are black, shiny, and pasty. Older droppings are brown and fall apart when touched. Figure 9.3 shows the features and droppings of Norway rats, roof rats, and house mice. Table 9.1 describes these three rodents.

Rats and mice usually follow the same path or runway between their nests and food sources. After awhile, grease and dirt from their bodies form visible streaks on floors and other surfaces. Rodents tend to stay close to vertical surfaces when they travel, so their tracks may go along walls, rafters, steps, or the sides of pipes. Rat and

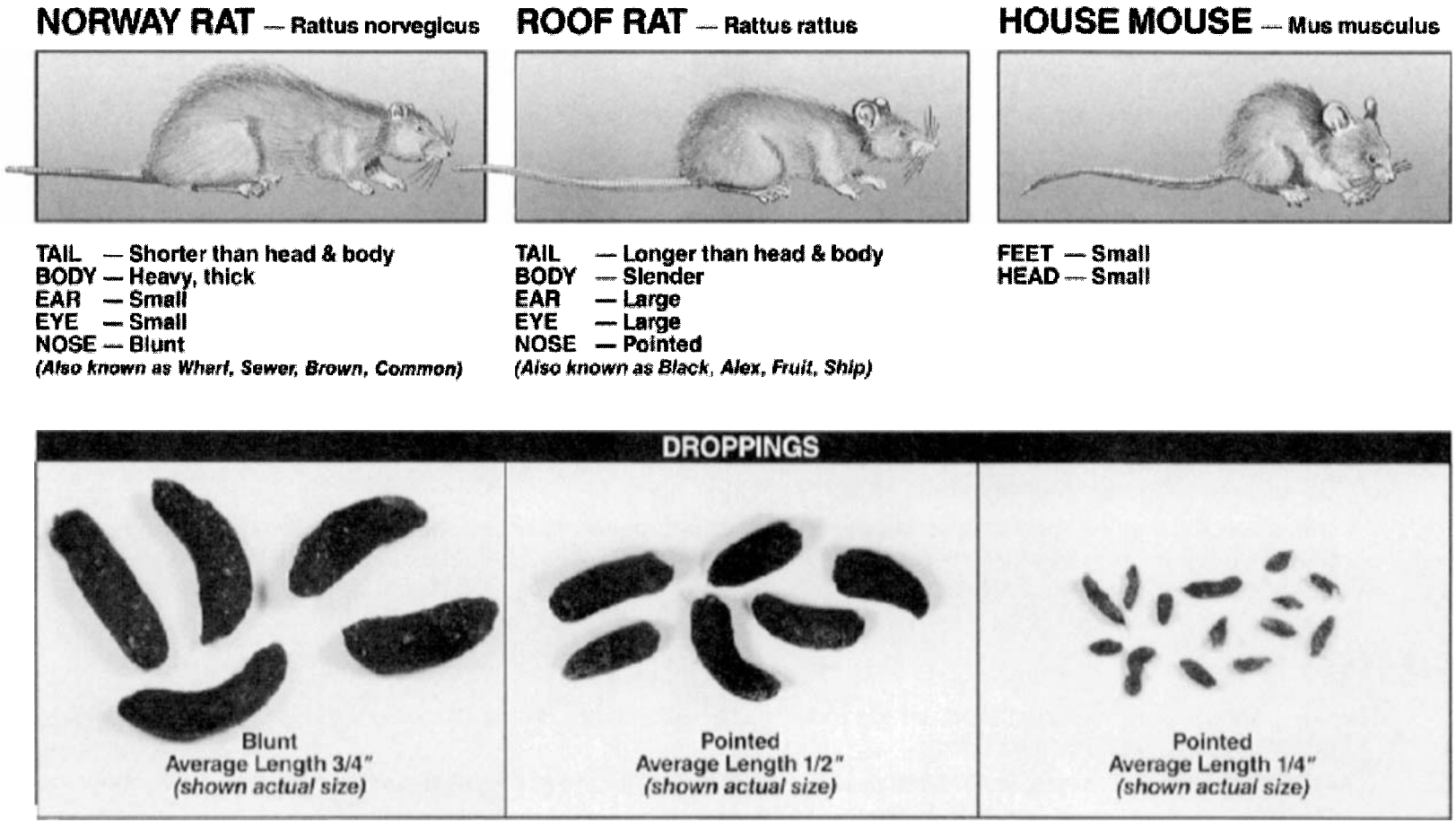


FIGURE 9.3. Appearance and droppings of common rats and mice.

TABLE 9.1. Facts About Common Rats and Mice

Name	Average Weight, oz	Body	Tail	Ears	Color	Food Preferences	Food Consumed Per Day	Droppings Per Day	Urine Per Day, cc	Habits
Norway rat *Rattus Norvegicus*	10–17	Heavy, broad, 7–10 in long, blunt head	6–8.5 in long, lighter color on underside	Average, close to body	Brown to black on back and sides, gray to yellow-white on belly	Meats, fish, flour, fruits, vegetables;. eats most human foods	3/4–1 oz food 1/2–1 oz water No. young in litter, 6–18	30–180 # litters per year 3–7	15.7 Length adult life, 18 mo	Usually nests in basements & lower portions of buildings. Burrows in soil extensively. Active primarily at night. Fair climber, good swimmer.
Roof rat *Rattus rattus Alexandrinus*	6–12	Slender body 6.5–8 in long, pointed muzzle	7.5–10 in long Uniform color, top and bottom	Large, prominent	Tawny back, grayish-white underparts	Seeds, fruits, vegetables, grain, eggs, etc.	1/2–1 oz food, up to 1 oz water No. young in litter, 6–14	30–180 # litters per year 3–7	15.7 Length adult life, 18 mo	Usually enters and nests in upper portions of buildings. May nest outside in trees (esp. palms), ivy, etc. Burrows very little. Excellent climber. Active at night.
House mouse *Mus musculus*	1/2–3/4	Average body 2.5–3.5 in long	3–4 in	Large, prominent	Dusky gray on back, lighter gray on belly	Meats, grains, cereals, seeds, fruits, vegetables; eats most types of human food	1/10 oz food, 1/20 oz water No. young in litter, 6	50 # litters per year 8	1.75 Length adult life, 15–18 mo	May nest in any portion of a building near food. Has an average range of 50 ft from nest. Active mostly at night. Nibbles small amounts of food frequently, rather than having large meals.

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mouse tracks are easy to see on dusty surfaces. Operators can spread talc in areas where they think rodents are active to make tracks more visible. Long-wavelength UV light can show up urine stains as a yellow fluorescence on burlap bags, and a pale, blue-white fluorescence on kraft paper.

Gnaw marks also show when rats are present. The incisor teeth of rats are strong enough to gnaw through metal pipes, unhardened concrete, sacks, wood, and corrugated materials to reach food. Teeth marks are visible if gnawings are recent. Bumping noise, shrill squeaks, fighting noises, or gnawing sounds at night are clues that rodents are living in the building.

Control

Control of rodents, especially rats, is difficult because they adapt to the environment so well. The best way to control rodents is by proper sanitation. If entrances are sealed off, these pests cannot get in, and if there is no debris to eat, they will not survive. If sanitation is not good, poisons and traps will only reduce the rodent population temporarily.

Prevention of entry. To ratproof an establishment, staff need to block all possible entrances. Poorly fitting doors and poor masonry around external pipes need flashing or metal or concrete covers or fillings. Vents, drains, and windows need screens. Decay in building foundations allows rats to burrow into buildings, so maintenance staff should keep masonry repaired and should block fan openings and other potential entrances.

It is easier to control rodents if they have no place to live. Outside equipment should be kept 23 to 30 cm (10–12 in) off the floor, shrubbery should be at least 10 m (30 ft) away from food facilities, and food-processing buildings are best surrounded by a grass-free strip 0.6 to 0.9 m (2–3 ft) wide covered with a layer of gravel or stones 2.5 to 3.8 cm (1–1.5 in) deep. This helps control weeds and rodents.

Eliminating rodent shelters. Crowded and unclean storage rooms provide sheltered areas where rodents can build nests and reproduce. Rodents love garbage and other refuse. Refuse is less attractive to rodents if it is stored 0.5 m (18 in) above the floor or ground. Storing waste containers on concrete blocks eliminates hiding places under the containers. Waste containers should be made of heavy-duty plastic or galvanized metal and have tight-fitting lids.

Foodstuffs should be stored on racks at least 15 cm (6 in) above the floor or away from the walls. A white strip painted around the edge of the floor in storage areas reminds workers to stack products away from the walls and helps show up rodent tracks, droppings, and hair.

Eliminating rodent food sources. Storing food and supplies properly and proper cleaning can help eliminate rodent food sources. Workers should clean up spills promptly, sweep floors regularly, and take out trash regularly. They should store food ingredients and supplies in tightly sealed containers.

Poisoning. Poisoning can be a good way to kill rodents, although the poison can also be dangerous to human beings. The most common rodenticides are fumarin, warfarin, and pival. (Warfarin is not used much in commercial establishments.) Rodents need to consume these poisons several times (multidose poisons) before they die, so adult humans who accidentally eat poisoned bait once are not in danger.

Multiple-dose poisons are safer than most other poisons, but workers still need to prepare and apply them according to the directions. Ideally, workers should put poison along rodent runways and near feeding sites. To make sure that humans do not accidentally eat the bait, workers should place it in bait boxes or under rodent shelters. Workers should put out fresh bait daily for at least 2 weeks to make sure that the poison works.

Rodenticides are commercially available in several forms. They are sold as ready-to-use baits to go in plastic or corrugated containers near rodent runways; in pellet form, mixed with grain to go in rodent burrows and dead spaces between walls; in small plastic packages to go in rodent hiding places; in bait blocks; and as salts to mix with water. The sanitarian or pest control operator should keep records of all bait containers so that they can be inspected and replaced easily. If rodents have not eaten the bait after two or more inspections, it should be moved.

If an operator needs to kill rodents immediately, he or she should use single-dose (acute) poisons such as red squill and zinc phosphide. Operators can mix these poisons with fresh bait, such as meat, cornmeal, or peanut butter. Operators should prepare and use these baits according to manufacturer's directions. Unfortunately, some of the single-dose poisons only work against Norway rats.

Sanitarians should place baits in several places, because rodents often travel only a short distance from their shelters. If there is enough food and shelter, rats stay within 50 m (150 ft) and mice within 10 m (30 ft) of their homes. If baits are too far apart or are in the wrong places, rodents may not find the poison. Several baits should be placed wherever there are signs of recent rodent activity. Because they often carry food to their nests, rodents may die there and workers may smell their bodies decomposing. Workers need to remove and burn or bury dead rodents.

Bait is one of the best ways to kill rodents, but rats that have been ill or in pain without dying after eating a poison may avoid the bait. They are also cautious if dead or dying rats are close to a bait. Sanitarians can overcome these problems using a nonpoisoned bait for about 1 week and then using the same bait with rodenticide. Prebaiting is very important with single-dose poisons but is not recommended with fumarin and warfarin. Prebaiting is not needed for mice, because their avoidance instincts are weaker.

Tracking powder. Toxic tracking powder kills rats, and nontoxic powder shows where rats are and how many there are. Toxic powders contain an anticoagulant (fumarin or warfarin) or a single-dose poison. The poison kills rodents when they groom themselves after running through the powder. Powders work well if the food supply is abundant and rodents are therefore less interested in bait. Tracking powders work better against mice than against rats.

Gassing. Operators should only use this technique if other eradication methods do not work. If necessary, a professional exterminator or a thoroughly trained employee should gas rodent burrows with a compound such as methyl bromide. They should not gas rodent burrows if they are less than 6 m (20 ft) from a building because the burrows may go under the building.

Trapping. This is a slow but generally safe method of killing rodents. Operators should place traps at right angles to rodent runways, with the baited or trigger end toward the wall. Any food that appeals to rodents works as bait. Operators should check traps daily, removing trapped rodents and replacing bait as necessary. Operators should only use traps with other methods of control and should use as many traps as possible. Rats are shy and adaptable and can avoid traps as well as bait. A glueboard, which traps a mouse's feet in glue, is a good mousetrap. The pest control operator throws away the disposable tray and trapped mouse and replaces it with a new tray.

Ultrasonic devices. Ultrasonic sound waves are supposed to keep rodents away. However, if rats are hungry, they ignore the sound. These are not effective rodent controls.

BIRDS

Birds such as pigeons, sparrows, and starlings can be a problem in food facilities. Their droppings are ugly and can carry microorganisms that cause disease in human beings. Birds can also carry insects into food establishments.

Proper management and sanitation can reduce the bird population. Birds are attracted by food waste, so proper waste disposal keeps them away. Screens on doors, windows, and ventilation openings help keep birds out.

Trapping is an acceptable way to control birds. Wires that give a mild electric shock and pastes that keep birds away also stop them from nesting near food establishments. But electric wires are expensive and need frequent inspection and maintenance. Flashing lights and noise have little effect on birds, because they soon get used to this equipment. Other techniques that work if they are kept up are removing bird nests and harassing birds by spraying them with water. The best way to get rid of birds is to use an exterminator who specializes in bird control. A professional exterminator has the proper expertise and equipment and can use the right chemicals safely.

Chemical poisons can reduce the number of birds, although these poisons are not safe inside a food establishment. Operators once used strychnine-coated cereal grains, but some local regulations now restrict use of strychnine. Operators should remove dead birds so that dogs and cats do not eat them and suffer from secondary poisoning. Another compound that controls the bird population is 4-aminopyridine. This compound kills birds and also makes the birds that eat it make distress sounds and behave abnormally, which frightens away other birds. Azacosterol makes birds temporarily sterile but is only approved for control of pigeons. This biological con-

trol method carries less risk than other compounds but only provides a long-term solution (in other words, it does not remove the problem immediately), especially in species like pigeons that have a long life. Temporary sterility is not a very valuable bird control for sanitarians who need to get rid of birds immediately.

Traps with live decoys are best for controlling birds. The biggest limitation of bird trapping is the cost of labor and materials.

USE OF PESTICIDES

Sanitarians should not spray insecticides in food areas while they are in operation; they should wait until after the shift, until the weekend, or other times when the food establishment is closed. They should be careful that the insecticide does not spatter or drift onto other surfaces or food. Insecticide dusts usually contain a dry form of the same toxic compounds in sprays. Dusts take more skill than sprays to apply. Only professional pest control operators should use them.

Before using insecticides, even those approved for edible food products or supply storage areas, workers should cover all exposed food and supply items or, better yet, remove them to another area. Sanitarians need to wash spraying equipment after using it by scrubbing with a cleaning compound and hot water and then rinsing. Sanitarians should not use products containing residual-type insecticides on any surfaces that food will touch. A fumigation procedure is not recommended unless other methods have not worked, and even then a professional fumigator should do it. Regular plant staff or supervisors should not try to fumigate the facility unless they have the right training. Even when professional fumigators are used, plant managers should make sure that they take all the right safety precautions

The following precautions are suggested by the National Restaurant Association (1992) when applying pesticides:

1. Make sure all pesticide containers have clear labels.
2. Check that exterminators have work insurance to protect the establishment, employees, and customers.
3. Follow instructions when using pesticides. Only use these chemicals for the purposes on the label. An insecticide that works against one type of insect may not work against other pests.
4. Use the weakest poison that will destroy the pests.
5. Use oil-based and water-based sprays in appropriate places. Use oil-based sprays where water could cause an electrical short circuit, shrink fabric, or cause mildew. Use water-based sprays in places where oil may cause fire, damage to rubber or asphalt, or a bad smell.
6. Avoid prolonged exposure to sprays. Wear protective clothing while applying pesticides, and wash hands afterwards.
7. Be careful not to contaminate food, equipment, and utensils with pesticides.
8. Call a doctor, poison control center, or fire department rescue squad if accidental poisoning occurs. If you cannot get immediate help, make the victim vomit by

putting a finger down the victim's throat. Then give the victim 2 tablespoons of Epsom salts or milk of magnesia in water, followed by one or more glasses of milk or water. If strychnine poisoning occurs, administer 1 tablespoon of salt in a glass of water within 10 minutes to induce vomiting, followed by 1 teaspoon of activated charcoal in half a glass of water. Lay the victim down, and keep the person warm.

Chemical pesticides cannot substitute for proper sanitation. Proper sanitary practices work better and are cheaper than pesticides. Even with good pesticides, pests keep coming back if conditions are not sanitary.

Food facilities can minimize contamination if they store only essential pesticides on the premises. Sanitarians should inspect pesticide supplies regularly to check the inventory amount and the condition of the chemical. Sanitarians should store pesticides carefully:

1. Store pesticides in a dry area at a temperature below 35°C (95°F).
2. Store pesticides in a locked cabinet away from food-handling and food storage areas. Store these compounds separately from other chemicals, such as cleaning compounds.
3. Do not transfer pesticides from their labeled package to any other storage container. Storage of pesticides in empty food containers can cause pesticide poisoning.
4. Place empty pesticide containers in plastic bins marked for disposal of hazardous wastes. Even empty containers can be dangerous because they still contain small amounts of toxins. Burn paper and cardboard, but do not burn empty aerosol cans. Follow local regulations about use and disposal of pesticides.

INTEGRATED PEST MANAGEMENT

Integrated pest management (IPM) uses several methods together to control pests. IPM aims to control pests inexpensively using environmentally sound techniques, including biological control. The goals of IPM are to use pesticides wisely and to find alternatives whenever possible. With IPM, operators can use less pesticide because other control methods are working at the same time.

The term IPM implies that pests are "managed" rather than removed. However, the final goal of pest management in food processing is to prevent or get rid of pests. Pest control practices include inspection, housekeeping, physical and mechanical methods, and chemical methods. Integrated use of these methods allows economical, effective, and safe pest management.

Inspection

Inspection is a preventive control measure. It is time consuming but important and cost effective in the long run. Inspection is a critical part of IPM. Inspection can find problems early on, pick up potential problems before they begin, and check that the sanitation program is working.

Sanitarians should carry out formal inspections at least monthly. The inspector should check thoroughly for signs of pests and evaluate how well the pest management program is working. If possible, well-qualified inspectors from outside the plant (e.g., corporate staff inspector, consultant, or contracting inspection service representative) should inspect the facility.

Plant personnel should carry out ongoing informal inspections in their own work areas. Supervisors should encourage employees to be aware of sanitation problems that make pest control less effective.

Inspectors should check raw materials, manufactured or prepared products, outside the building, inside facilities, and equipment. Inspectors should use a flashlight and have the correct tools to open equipment and sample containers. Sanitarians should put together an inspection form to make sure the inspector does not miss anything and to record results. The forms give a record of potential problems and how problems are solved.

Housekeeping

Employees need to understand and be responsible for maintaining the facility's standards of cleanliness and cleaning schedules. Cleaning must be continuous in many areas, because even small amounts of product residues can attract pests, and undisturbed spaces can provide homes for pests.

Physical and Mechanical Methods

Because many pesticides that used to be common can no longer be used for pest control, physical and mechanical methods are important. Examples are rodent traps, glue boards, and electric fly traps. These methods do not usually contaminate the area and can make an IPM program more effective overall. Use of very hot or cold temperatures, sometimes combined with forced air movement, can help control pests.

Insects need the right amount of moisture. Foods with a low moisture content (especially below 12%) do not usually support growth of insects.

Different forms of radiation (radio frequencies, microwaves, infrared and ultraviolet light, gamma rays, X rays, and accelerated electrons) can disinfect food products, but not all of these methods are effective and practical. Commercial food operations sometimes use gamma rays, X rays, and accelerated electrons to get rid of insects.

Chemical Methods

Pesticides and other chemicals such as repellents, pheromones, and sticky materials are also an important part of IPM. However, IPM uses fewer chemicals because this is not the only method used. Staff who apply pesticides must know how to use each chemical so that it is safe, legal, and effective.

The Environmental Protection Agency (EPA) classifies pesticides as general use and restricted use. Those classified as restricted use are more likely to damage the en-

vironment or hurt the person using them. Therefore, only certified applicators can purchase and use these pesticides. States offer an EPA-approved training and certification program for staff who will need to use these chemicals.

The pesticide storage area should be large enough to store normal supplies of pesticide materials neatly and safely. The pesticide storage area should be located away from food, in a separate building if possible. This storage area should have power ventilation exhausting to the outside and should never be cross-ventilated with food-processing or food container storage areas. This storage area should be totally enclosed by walls and locked, so that only authorized staff have access to pesticides. The storage area should be dry and at the right temperature to prevent the chemicals from breaking down or reacting. Pesticide containers should have a visible label, and the sanitarian should check the inventory regularly. Pesticide handling and application equipment should include rubber gloves, protective outer garments, and respirators such as dust masks or a self-contained breathing apparatus (SCUBA).

SUMMARY

- The pests that cause the most problems in the food industry include the German cockroach, American cockroach, Oriental cockroach, housefly, fruit fly, Norway rat, house mouse, pigeon, sparrow, and starling.
- The most important parts of pest control are preventing pests from getting into food establishments and getting rid of their shelter areas and food sources.
- If pests become established, pesticides, traps, and other controls are essential.
- Pesticides and traps can never replace good sanitation practices.
- Pesticides are toxic, so sanitarians should select, store, use, and dispose of them carefully. A trained employee can handle pesticides, but it is better to employ a professional exterminator for dangerous chemicals.

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STUDY QUESTIONS

1. Which insects cause most problems in food establishments?
2. How do pests spread disease?
3. How would you find out if a facility is infested with cockroaches?
4. How can you prevent cockroach infestations?
5. What are the best ways to control flies?
6. What is the difference between residual and nonresidual pesticides?
7. List four ways to control rodents.
8. List four precautions when using pesticides.
9. What is integrated pest management?

TO FIND OUT MORE ABOUT PEST CONTROL

1. Contact the National Pest Control Association, (1-800) 678-6722, and ask for information.
2. Contact a sanitation and pest control products and services company (e.g., Diversy Corporation, (1-313) 458-5000 or Ecolab Inc., (1-612) 293-2233), and ask for information on pest control.
3. Use the Yellow Pages to find local pest control services. Are there any who specialize in pest control in food establishments? Call and ask them about the problems they encounter most often in your area.
4. If you work in food production or foodservice, ask your employer what is done to control pests at your facility.



COCKROACHES AND BACTERIA

People are often afraid of cockroaches and consider them dirty. They also carry an unsanitary collection of bacteria, fungi, helminths, and viruses. To find out how likely cockroaches are to cause disease, the authors of this study collected cockroaches from several places to find out which bacteria they were carrying. They collected cockroaches from a hospital nursing area and out-patient area, a swimming pool pool-side and toilet area, low-income apartments, and commercial kitchens. The 157 samples (each sample contained 5 to 10 cockroaches) contained 56 types of bacteria, 14 of them known to cause disease in humans. Cockroaches collected from the swimming pool pool-side carried fewer bacteria, probably because of the chlorine in the water. The authors conclude that cockroaches can be extremely hazardous to human health and that pest management programs are essential, especially in commercial kitchens and hospitals.

Source: Rivault, C., Cloarec, A., and Le Guyader, A. 1993. Bacterial load of cockroaches in relation to urban environment. *Epidemiol. Infect.* 110:317.